



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

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CIVICS GROUP

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H2 BIOLOGY

Paper 3

Long Structured Question and Free-response Questions

9744/03

14 September 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/ 25
2	/ 13
3	/ 12
Section B	/ 25
Total	/ 75

Section A

Answer **all** the questions in this section.

- 1 The arginine (*arg*) operon in *E. coli* is a repressible system with three genes, *arg B*, *arg C* and *arg H*, that code for enzymes responsible for the synthesis of the amino acid arginine. The *arg* operon is shown in Fig. 1.1.

Another gene upstream of the operon (not shown in Fig. 1.1) codes for a repressor protein R. It binds to the operator to regulate the operon. Arginine binds to repressor R and activates it.

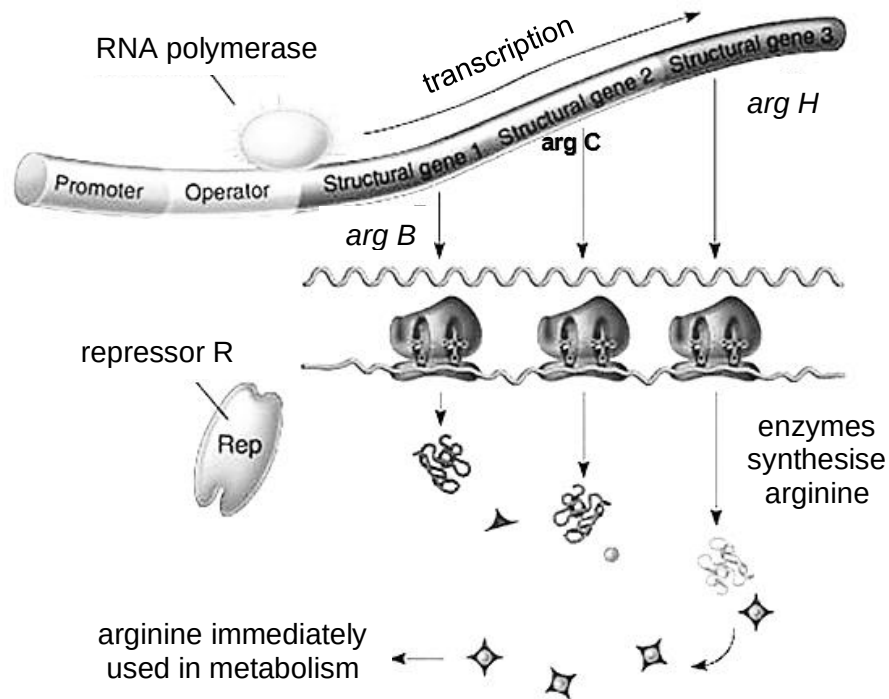


Fig. 1.1

- (a) (i) Explain why the *arg* operon is a repressible system. [2]

1. It is usually expressed, but it can be switched off / repressed
2. when arginine acting as a co-repressor binds to the repressor R protein which then binds to the operator of the operon and prevents transcription of the structural genes.

- (ii) Describe one similarity and one difference between the repressible system shown by *arg* operon and the end product inhibition of enzyme activity in bacteria. [2]

Any 1 similarity:

- Both work to conserve resources of the bacterial cell by limiting the production of a molecule when there is excess of the molecule.
- The (excess) products are involved in their own inhibition.
- Both are self-regulatory. As the molecule is used up (or needed), the inhibition is lifted and its production is switched back on again.

Any 1 difference:

- Repressible system inhibits the expression of the genes whereas end product inhibition inhibits the activity of an enzyme involved in the early stages of the metabolic pathway.
- Repressible system involves activation of a repressor but end production inhibition involves inhibition of an enzyme.
- Repressible system involves co-repressor bind and activate a repressor but end production inhibition involves an inhibitor bind and inhibit an enzyme.

Mutations were introduced into *E. coli* strains **E1**, **E2** and **E3**. The activity of *arg B* gene, in the form of presence or absence of enzyme activity, was measured in different *E. coli* strains.

The results are shown in Table 1.1.

Table 1.1

<i>E. coli</i> strain	genotype	presence (✓) or absence (✗) of enzyme activity	
		arginine present	arginine absent
E1	R ⁺ O ⁺ B ⁻	✗	✗
E2	R ⁻ O ⁺ B ⁺	✓	✓
E3	R ⁺ O ⁻ B ⁺	✗	✗

Key: R⁺ – normal repressor gene *R*
 R⁻ – mutated repressor gene *R*

 O⁺ – normal operator
 O⁻ – mutated operator

 B⁺ – normal *arg B* gene
 B⁻ – mutated *arg B* gene

- (b) With reference to Table 1.1, explain the abnormal *arg B* gene activity in strains **E2** and **E3**. [3]

[Strain E2]

1. Mutated repressor gene *R* codes for a non-functional repressor protein.
2. Even in the presence of arginine, repressor does not bind to arginine.
3. *Arg B* gene is transcribed resulting in enzyme activity.

[Strain E3]

4. Operator is mutated. Even in the presence of arginine, the (inactive) repressor protein still binds to the operator and blocks RNA polymerase from binding to the promoter.
5. No transcription of *arg B* gene resulting in no enzyme activity.

- (c) Bacteria may be exposed to different types of stress, and hence had developed appropriate responses in order to survive. A variety of heat shock proteins (HSP) that are expressed in response to changes in temperature have been identified from *E. coli*.

To study the expression levels of the different HSP genes, bacterial cells were subjected to heat stress of 45°C. Proteins were then extracted from *E. coli* and separated by two-dimensional (2D) gel electrophoresis. This technique is carried out in a sequence shown in Fig. 1.2.

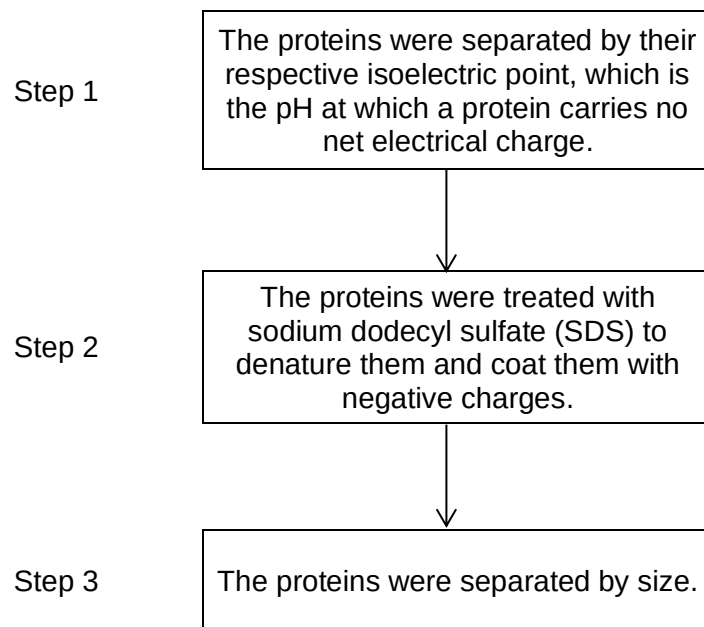


Fig. 1.2

- (i) Suggest why SDS treatment is required in step 2.

1. SDS is required to linearise the proteins (denaturation) so that they may be separated by molecular size alone and not due to shape.

OR

2. SDS coats the proteins with negative charges so that they will migrate toward the positive electrode / anode.

The result of the 2D gel electrophoresis is shown in Fig.1.3. The size and intensity of each protein spot on the gel corresponds to its amount.

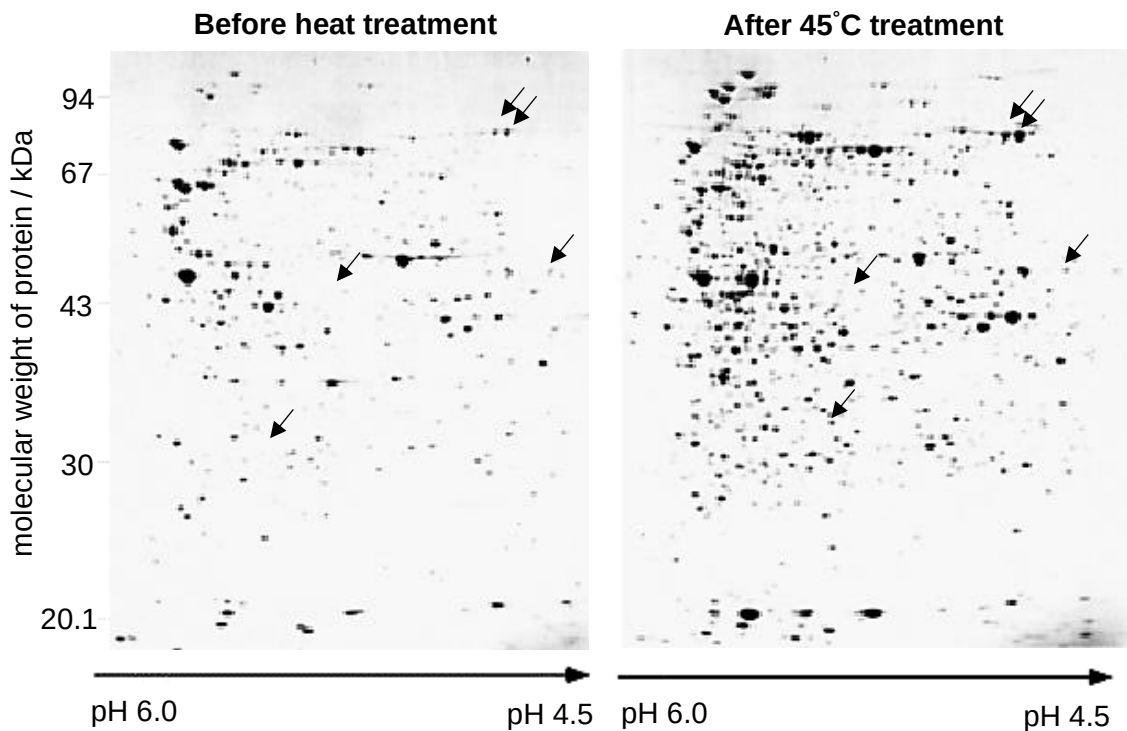


Fig. 1.3

The experiment was carried out six times and the expression levels of five types of HSP (arrowed) before and after heat treatment is shown in Fig 1.4.

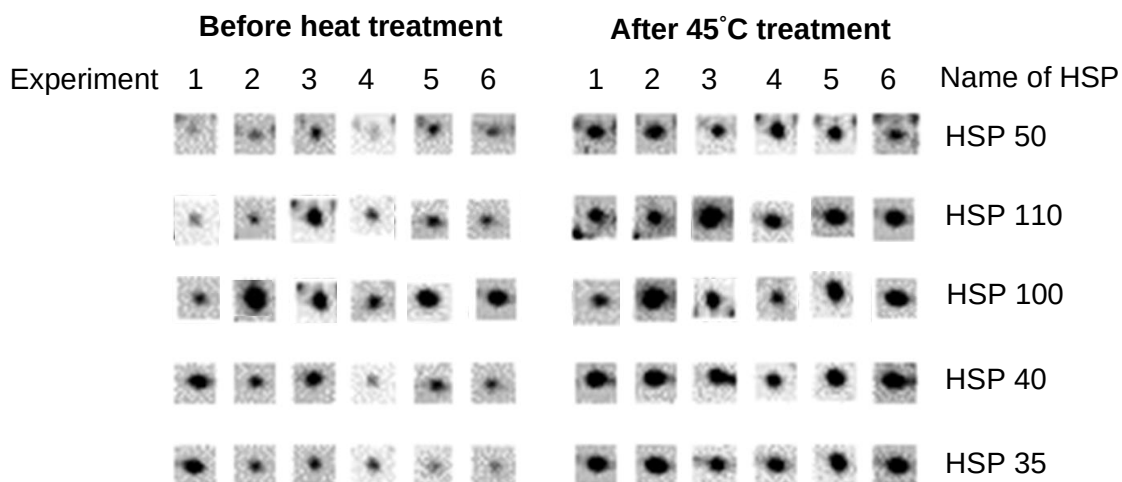


Fig. 1.4

(ii) Using the information provided, deduce the identity of the heat shock protein that did not respond to the heat treatment. [1]

- **HSP 100**

(iii) Give a reason for your answer in (c)(ii). [1]

The size and intensity of the spots were similar before and after heat treatment.

Using 2D gel electrophoresis, the expression levels of different genes of an *E. coli* strain at different temperatures were determined and shown in Fig. 1.5.

Each data point represents an individual gene.

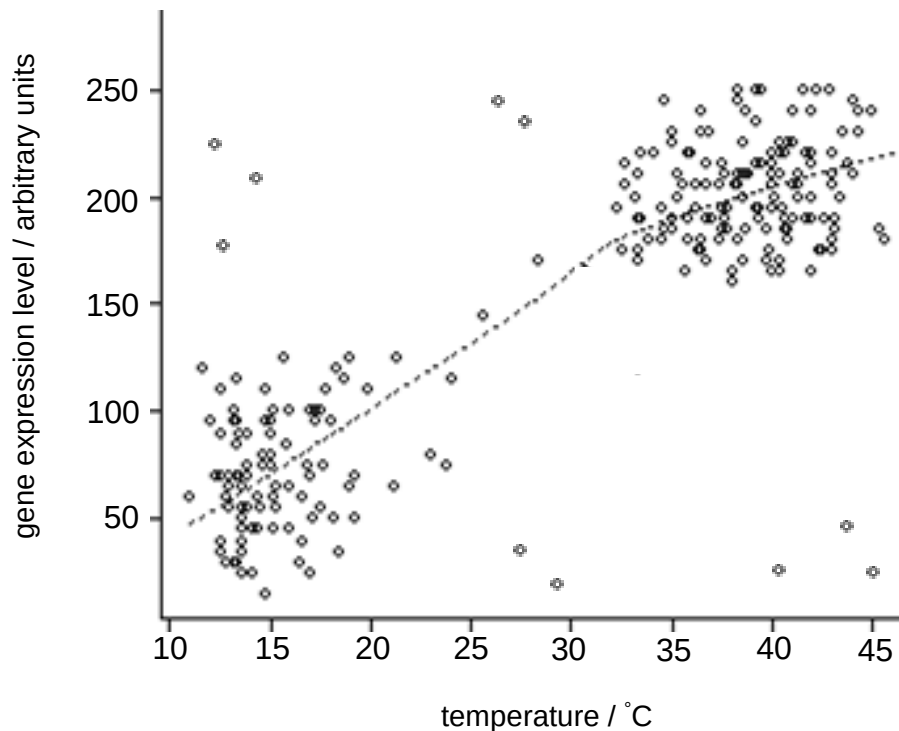


Fig. 1.5

(iv) With reference to Fig. 1.5, describe the extent to which temperature affects the expression levels of the genes. [3]

1. In general, higher temperature increases the level of gene expression.
2. As temperature increases from 11°C to 45°C, gene expression increases from (accept: 50 – 60 au) to (accept: 170 - 250 au)
3. At lower temperatures, there are many genes expressed
4. Temperature range of 11 – 20°C, low expression 5 – 130 au.
5. At Intermediate temperatures, a few genes expressed
6. Intermediate temperatures 20°C – 32°C, wide range of expression level 25 – 250 au.
7. At higher temperatures, there are many genes expressed
8. Temperature range of 32 – 45°C, high expression 150 – 250 au.

However, there are deviation from the trend.

9. At lower temperatures, a few genes are expressed at high level
10. Low temperatures, ranging 11 – 20°C, high expression 175 – 225 au
11. At higher temperatures, a few genes are expressed at low level
12. High temperatures, ranging 32.5 – 45°C, low expression 25 – 50 au

HSP 35 is known to be a protease. It specifically hydrolyses the peptide bond between phenylalanine (Phe) and alanine (Ala). Thus, a small tripeptide consisting of glycine, phenylalanine and alanine, Gly-Phe-Ala, is hydrolysed to Gly-Phe and Ala.

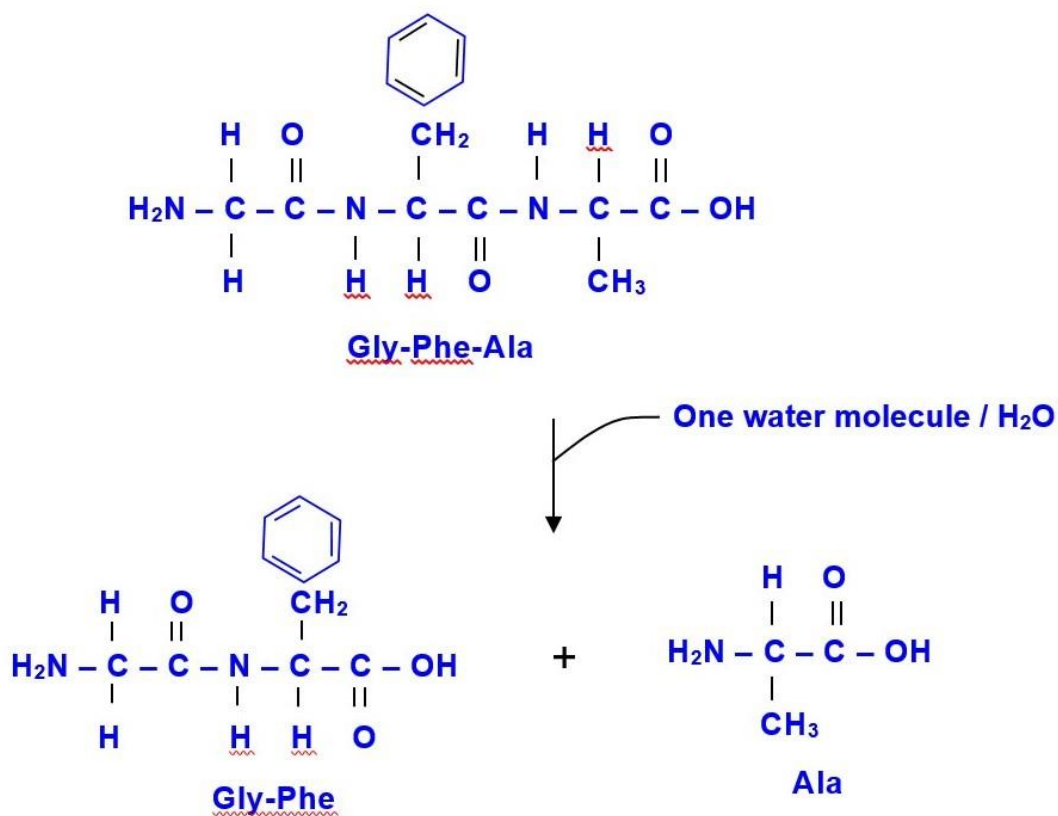
- (v) In the space below, draw a labelled diagram to illustrate the above enzymatic reaction. Include the respective R groups of the amino acids in your answer.

Legend:

R group of Gly: $-H$

R group of Ala: $-CH_3$

R group of Phe: $-CH_2$ 



[3]

- (d) Scientists produced genetically engineered bacterial cells. The gene coding for the green fluorescent protein (GFP) is joined upstream of the heat-shock gene.

Promoter	Operator	GFP gene	Heat-shock gene
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Fig. 1.6

Both genes are transcribed to produce a single mRNA molecule. When translated, a “fused” protein is produced.

- (i) Describe one feature of the control of eukaryotic transcription that is **not** shown in Fig. 1.6. [2]

1. **Enhancer is absent.**
2. **Enhancer can be found upstream or downstream of the eukaryotic gene.**

OR

3. **Enhancer allows the binding of activator to bend DNA**
4. **which stabilises the formation of transcription initiation complex to increase the rate of gene transcription.**

The “fused” protein shows up as a green fluorescent spot when viewed using a fluorescent microscope. The amount of the “fused” protein may be determined by the intensity of the green fluorescence.

- (ii) Suggest one advantage of this method over 2D gel electrophoresis in determining the amount of protein. [1]

Any 1:

1. **It can determine the relative amount of protein without killing the cell. 2D gel electrophoresis involves extracting proteins from cells by lysing and hence killing the cells.**

OR

2. **It can detect the relative amount of proteins in specific locations in the cell and so can also derive valuable information regarding the likely function of the protein. For example, detecting most of the fluorescence in the nucleus may suggest that the protein is a transcription factor.**

- (e) Besides having operons, bacteria have other means to adapt to changing environments through horizontal gene transfer. A method is through bacterial transformation.

- (i) Outline the process of bacterial transformation. [2]

1. **DNA-binding complexes on the outer surface of the (competent) bacterial cell binds to foreign DNA from external / surrounding environment.**
2. **After binding, DNA moves across the cell surface membrane.**
3. **At the same time, bacterial nucleases degrade one of the strands.**
4. **The single-stranded DNA fragment can then integrate into the bacterial chromosome.**

Insulin injection is required by diabetic patients on a daily basis. To produce sufficient insulin to meet the global demand, scientists inserted the human insulin gene into a bacterial plasmid DNA to form a recombinant plasmid. This gene was derived by reverse transcribing the mature mRNA.

The recombinant plasmid is introduced into bacterial cells by transformation that involves a chemical method. Upon expression of the human insulin gene, the insulin protein is then extracted from the bacterial cells and purified.

The production of human insulin using bacteria is shown in Fig. 1.7.

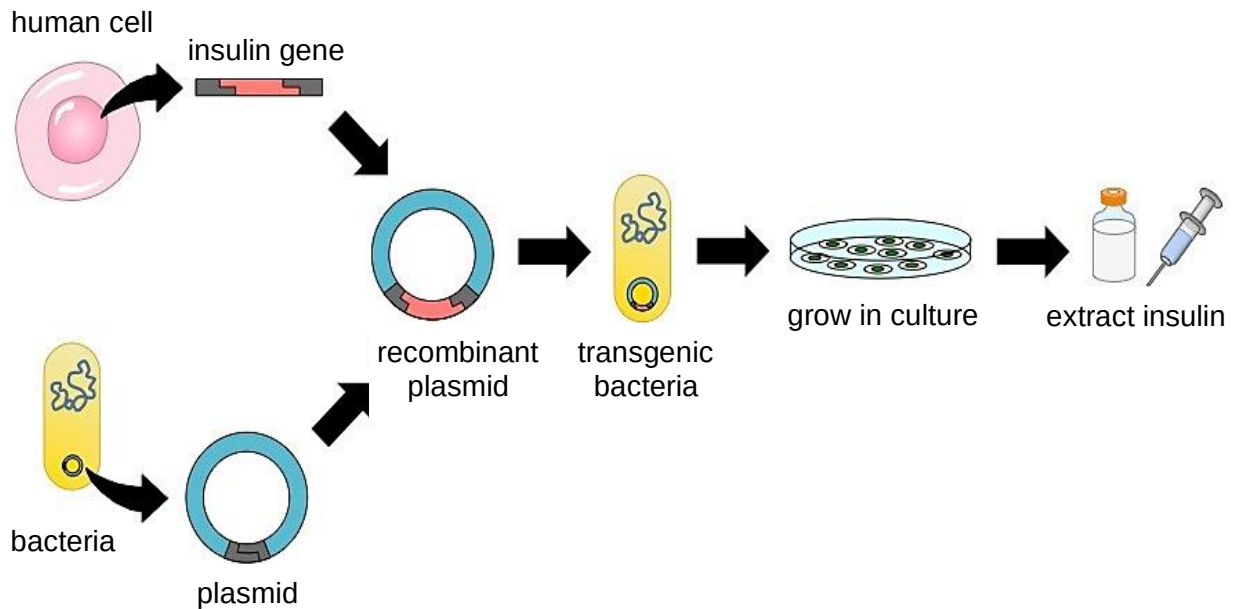


Fig. 1.7

(ii) Suggest why bacteria is used for the production of human insulin. [1]

1. Bacteria are easy/cheap to culture
2. Bacteria multiply quickly

Point 1 or 2 + idea of mass production / large amount of (human) insulin

A young scientist wanted to synthesise human insulin in his own laboratory. He extracted the human insulin gene successfully. Before the human insulin gene was inserted into the plasmid, he removed its promoter.

He then inserted the gene at a site in a plasmid that is just downstream of a *lac* operon promoter which he had also inserted into the plasmid.

- (iii) Explain why he decided to use *lac* operon promoter instead of relying on the promoter of the human insulin gene. [1]

1. **Bacteria are prokaryotes.**
2. **The (prokaryotic) RNA polymerase in bacteria would only be able to recognise and bind to the prokaryotic promoter and transcribe the (eukaryotic) gene.**

- (iv) The young scientist followed the steps shown in Fig. 1.7 carefully. However, the gene used was extracted directly from nuclear DNA.

Suggest why the scientist is unlikely to obtain functional insulin even if the gene was transcribed successfully. [2]

1. **The bacteria are unable to remove the introns of the insulin mRNA because prokaryotes do not undergo mRNA processing.**
2. **The introns would be translated resulting in the insulin protein with a different (longer) amino acid sequence. This affected the folding of the polypeptide resulting in a non-functional protein.**

AVP:

Bacteria unable to carry out post-translational modification by cleaving the pro-insulin to form the functional insulin.

[Total: 25]

- 2 The Hawaiian Islands are ordered by age, with the oldest islands in the North-West and the youngest in the South-East.

Fig. 2.1 shows the main Hawaiian Islands together with their date of formation, millions of years (My) ago.

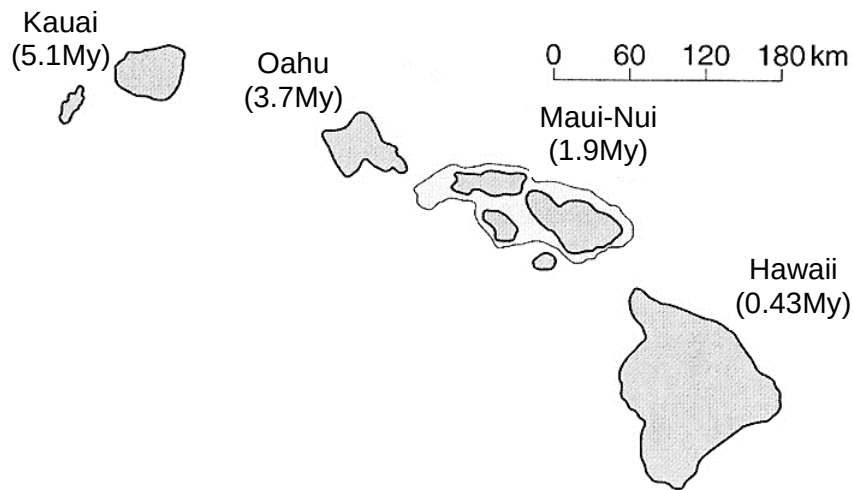


Fig. 2.1

Three closely related species of honeycreeper (a type of bird) are found on three different islands.

- Oahu Creeper, *Hemignathus kauaiensis*
- Maui Creeper, *Hemignathus flavus*
- Hawaii Creeper, *Hemignathus virens*

The date of an island's formation provides a maximum age for species inhabiting the island, assuming that populations colonise soon after the island's emergence.

- (a) With reference to Fig. 2.1 and the information given, explain how islands, such as the Hawaiian Islands, favour the formation of new species. [6]

1. Islands promote speciation through adaptive radiation.
2. Islands provide many unoccupied ecological niches as it takes time for populations to colonise after the islands' emergence.
3. Different islands / different parts of an island have different habitats, and hence, different environmental conditions.
4. Different selection pressures are present.
5. Organisms with favourable phenotypes/traits are able to adapt / be selected for.
6. They survive and reproduce to produce viable and fertile offspring, passing on their favourable alleles to their offspring.
7. Few competitors and predators are present as it is difficult for these organisms to reach the islands.
8. Well-adapted organisms can easily colonise the islands by reproducing at a high rate, thus rapidly diversifying to fill all the ecological niches.
9. The different populations are usually geographically isolated from a different island due to the sea.
10. E.g. Populations in Hawaii, Maui-Nui, Oahu and Kauai are geographically isolated.
11. Individuals from different populations / islands are not able to interbreed with each other
- OR
- absence / disruption of gene flow.
12. This promotes allopatric speciation.
13. Different habitats with different selection pressures in the same island could also promote the development of behavioural / physiological isolation.
14. This promotes sympatric speciation.
15. Isolated populations undergo evolution as the agents of evolution / mutation / genetic drift operate independently.
16. Over time, their isolated gene pools diverge, resulting in the eventual formation of new species.
17. The pioneers of the newly isolated population are not likely to carry all the alleles present in the original population / only represent a small percentage of the original population's gene pool.
18. Genetic drift due to Founder effect occurred, leading to a change in allele frequency and possible development of a new species.

- (b) Differences in the DNA sequences that encode cytochrome b within the individual species have been measured. Cytochrome b is a protein found in the electron transport chain.

Fig. 2.2 shows differences in cytochrome b DNA within the individual species plotted against time since colonisation (My).

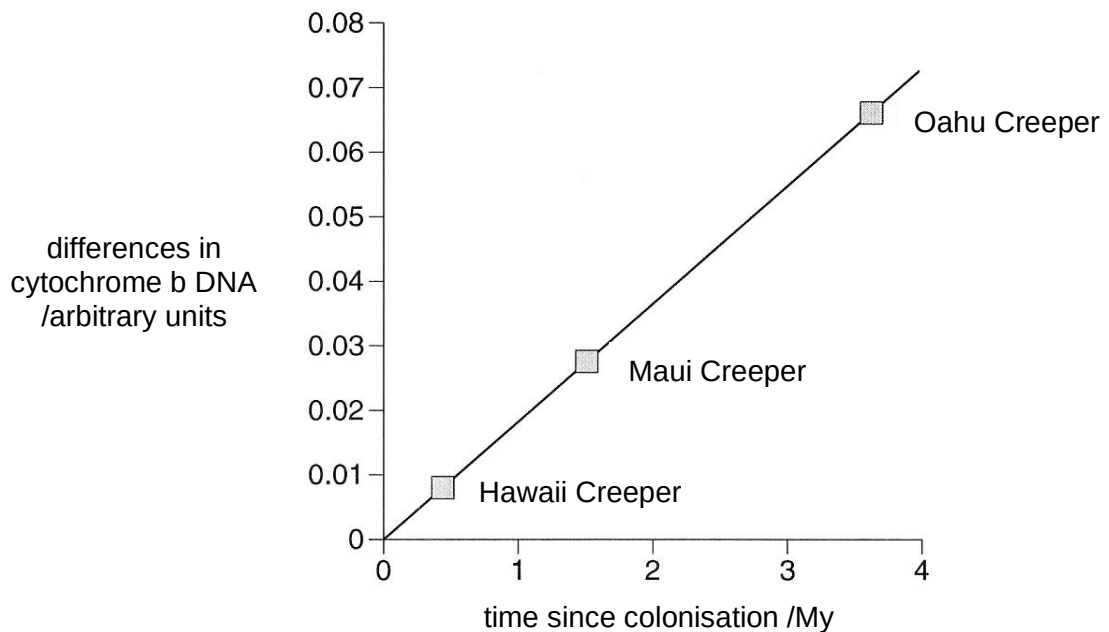


Fig. 2.2

- (iv) Suggest why the cytochrome b gene is used to measure changes in DNA sequences in closely related species. [1]

1. Found in all aerobic eukaryotes / all of the species. Hence, it's a good basis for comparison.

OR

- The gene is located in the mitochondrial DNA.
- Hence, it's not subjected to effects of crossing over / little or no genetic recombination / higher mutation rate than chromosomal DNA / passed down maternal line only.

- (v) Explain the trend in the differences in cytochrome b DNA within the individual species. [2]

- As the time since colonisation increases, the differences in cytochrome b DNA within the individual species increases.
- More time for mutations to occur within individual species as time since colonisation increases, hence more differences in cytochrome b DNA observed within the individual species.

- (c) Fossils provide strong evidence for Darwin's theory of evolution.

A fossil named *Ambulocetus* was discovered in 1992 in Pakistan, and it was dated at roughly 47 – 48 million years old. Paleontologists (scientists who study the history of life on Earth through fossil records) believe that this fossil is a transitional fossil between ancestral land mammals and present-day cetaceans (aquatic mammals, e.g. dolphins, whales).

Fig. 2.3 shows an artist's impression of *Ambulocetus*.



Fig. 2.3

- (i) With reference to Fig. 2.3, describe two structural features of the *Ambulocetus* that would support the hypothesis that it is a transitional fossil. [2]

[Feature that shows adaptation to land]

1. Four limbs / fore and hind limbs for walking / waddling/locomotion on land (like a sea lion)

[Feature that shows adaptation to water]

2. Webbed feet / flippers for swimming/propelling forward in water.

OR

3. Long snout and eyes near the top of the head for ambushing prey in water (like a crocodile).

OR

4. Streamlined body for swimming.

- (ii) Explain why transitional fossils provide strong evidence for Darwin's theory of evolution. [2]

1. Transitional fossils reveal the intermediate links between an ancestral group and its descendants.

2.

This is important especially when the ancestral group and its descendant groups are highly different / accumulations of modifications are not apparent, as the theory of evolution is that of descent with modification.

[Total: 13]

- 3 Fig. 3.1 shows the seaweed *Laminaria hyperborea*. This is a photosynthetic organism found in the coastal waters around Norway.

The seaweed is grown commercially to obtain the polysaccharide, alginate. Alginate is made up of repeating units of glucose. This is used in certain food products.



Fig. 3.1

An increase in carbon dioxide concentration in the atmosphere has resulted in higher concentrations of carbon dioxide in the ocean. This has caused a decrease in the pH of the ocean and has resulted in ocean acidification.

Scientists are studying seaweeds such as *L. hyperborea* because they absorb a large quantity of carbon dioxide during photosynthesis. This may help to increase the pH of the ocean and reverse ocean acidification.

- (a) Outline the reactions occurring in the stroma of the chloroplast that lead to the production of a polysaccharide, such as alginate. [4]
1. Calvin cycle occurs in the stroma.
 2. Rubisco catalyses carbon fixation.
 3. CO₂ combines with ribulose-1,5-bisphosphate (RuBP) to form an unstable 6C compound,
 4. which breaks down into 2 molecules of glycerate-3-phosphate (PGA / GP).
 5. Glycerate-3-phosphate is phosphorylated and then reduced to glyceraldehyde-3-phosphate (TP / GALP),
 6. using ATP and NADPH from light-dependent reaction.
 7. GALP / TP can be used to form glucose
 8. which can then be used to form alginate via condensation reaction / formation of glycosidic bonds between glucose monomers.

(b) Laboratory experiments were carried out to investigate the effect of day length on the rate of photosynthesis in another marine organism, *Zostera marina*.

- The temperature was controlled at 4 °C.
- A low concentration of carbon dioxide dissolved in the water was used.
- The light exposure period (day length) was different for five groups of *Z. marina*.
- This was maintained for 10 days to allow *Z. marina* to adapt to these conditions.
- After 10 days, the rate of photosynthesis was measured for each group under the **same** controlled conditions.
- The experiment was repeated using five groups of *Z. marina* with a high concentration of carbon dioxide dissolved in water.

Table 3.1 shows the rate of photosynthesis for each group.

Table 3.1

day length / hours	rate of photosynthesis / arbitrary units	
	low carbon dioxide concentration	high carbon dioxide concentration
12	2.0	2.5
14	3.0	5.0
16	4.0	7.0
18	5.5	11.0
20	7.5	18.0

(i) With reference to Table 3.1, explain the difference in the rate of photosynthesis at high carbon dioxide concentration compared to low carbon dioxide concentration. [1]

1. There is more Calvin cycle / light-independent reaction where
2. more CO₂ reacts with RuBP / more CO₂ fixation.

(ii) With reference to Table 3.1, describe **and** explain the effect of increasing day length on the rate of photosynthesis for the *Z. marina* in high carbon dioxide concentration. [4]

1. As day length increases from 12 hours to 20 hours, the rate of photosynthesis increases from 2.5 a.u. to 18 a.u.
2. This is because as day length increases, light intensity also increases.
3. More light energy is absorbed by photosynthetic pigments / chlorophylls,
4. leading to more photophosphorylation
5. and more excited electrons are passed down electron carriers of decreasing energy levels in the electron transport chain.
6. There is more photolysis of water to replenish displaced electrons.
7. Thus, more oxygen / reduced NADP / ATP / glucose produced.

- (c) In the laboratory, a seaweed was grown in water with different pH values. All other variables, including temperature and light, were standardised.

The mean rate of photosynthesis was calculated over a 24 hour period for each pH value.

The results are shown in Fig. 3.2.

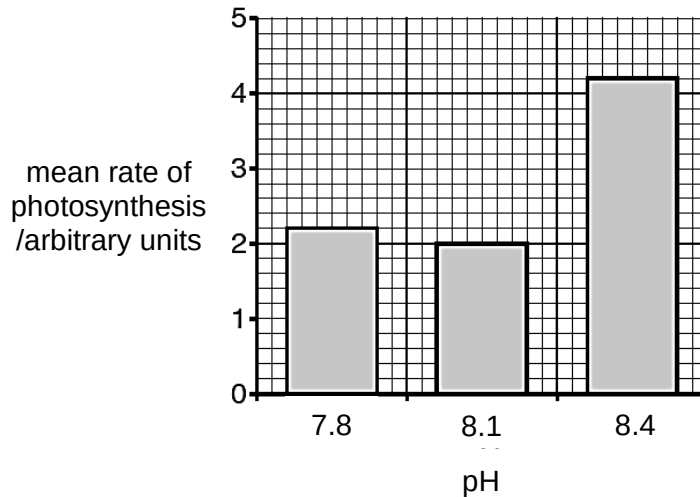


Fig. 3.2

- (i) With reference to Fig. 3.2, explain the effect on the rate of photosynthesis when the pH increases from 8.1 to 8.4. [2]

1. pH 8.4 could be the optimum pH of Rubisco / photosynthetic enzymes.
2. Thus, the lower H^+ concentration leads to a more suitable 3D conformation of the enzyme / active site.
3. **Idea of** greater effective collisions between enzyme and substrate / more enzyme-substrate complexes formed
4. Hence resulting in higher mean rate of photosynthesis at 4.2 a.u.

- (ii) The lower pH values on Fig. 3.2 represent ocean acidification.

Suggest why the results for the lower pH values do **not** fully support the idea that seaweeds can help to reduce ocean acidification. [1]

1. Less carbon dioxide is absorbed / fixed by the seaweeds, so ocean acidification is not reduced.

[Total: 12]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)** as indicated in the question.

- 4 (a) Outline the structures and functions of named organelles involved in the synthesis of polypeptides in eukaryotes. [10]

1. Nucleus

[Structure]

2. Spherical or ovoid structure, about 10–20µm in diameter.
3. Surrounded by nuclear envelope.
4. Nuclear envelope is perforated with nuclear pores.
5. Nucleoplasm is a semi-fluid matrix containing DNA in the form of chromatin.
6. Nucleoplasm contains enzymes required for transcription / post-transcriptional modification.
7. Consists of dense structure known as nucleolus.
8. Nucleolus contains the rRNA genes (DNA) that code for ribosomal RNA (rRNA).

[Function]

9. Contain and store hereditary material (DNA).
10. Control cellular activities by regulating the rate of protein synthesis (transcription).
11. Nuclear pore complexes control the movement of water-soluble substances between the nucleus and cytoplasm.
12. Site of synthesis of ribosomal RNA and partial assembly of ribosomes.

13. Ribosome

14. Free ribosomes.
15. Bound on rough endoplasmic reticulum (rER)

[Structure]

16. 20–25nm in diameter.
17. Each ribosome consists of 2 subunits: large and small ribosomal subunit.
18. Each subunit composed of rRNA and ribosomal protein.
19. rER: an extensive network of interconnecting membranous tubules and flattened sacs called cisternae,
20. continuous with outer membrane of nuclear envelope.

[Function]

21. Ribosomes attached provides the site of protein synthesis / surface of rER has receptor sites for attachment of ribosomes for translation of mRNA.
22. Polypeptide synthesised by ribosome passes through channel of the receptor protein into the lumen of the rER.
23. Packaging of proteins in ER vesicles for transport to the Golgi Apparatus.

- (b) Smallpox is an infectious disease that was eradicated in the 20th century with the aid of a successful global vaccination effort. However, there remains many other infectious diseases that could not be eradicated yet, and the human immune system is always in a constant race to produce a diversity of antibodies against these known and unknown pathogens.

Discuss the factors that led to a successful eradication effort against smallpox disease and describe how the immune system is able to rapidly and strongly respond to a wide variety of pathogens through the production of antibodies.

[15]

1. Herd immunity was achieved as a significantly large proportion of a population was vaccinated and developed immunity against smallpox.
2. With fewer susceptible individuals, this broke the disease transmission cycle and reduced spread of disease.
3. Infected individuals were easy to identify based on the symptoms of the disease.
4. The virus did not rapidly mutate or change its surface antigens.
5. The vaccine was effective in stimulating a strong immune response without causing disease.
6. The vaccine was cheap and easy to mass produce.
7. *Idea of* longer shelf-life.
8. The vaccine can be effective even with a small dosage.
9. There is secondary immune response which refers to the immune response elicited by the body upon re-exposure to the same antigen.
10. It is mediated by memory T cells and memory B cells.
11. Activated CD4 memory T cell proliferates and differentiates to helper T cells and memory cells.
12. Memory B cell recognise and binds to same pathogen and cytokines from helper T cells completes activation of memory B cell.
13. B cells quickly proliferate and differentiate to form plasma cells which produce larger amount of IgG antibodies specific to the antigen and also form memory cells.
14. Memory cells help to provide a rapid and stronger immune response to combat the infection by the same pathogen in the future.
15. Somatic recombination
16. where there can be different combinations of V and J gene segments coding for the variable region of the light chain
17. where there can be different combinations of V, D and J gene segments coding for the variable region of the heavy chain.
18. V, D, J gene segments not selected are permanently deleted.
19. This leads to different antigen-binding specificity in the variable region.
20. Somatic hypermutation
21. where high rates of mutations occur in both heavy chain and light chain variable region coding sequences.
22. This results in affinity maturation.
23. B cells can switch from the one antibody class to another (e.g. IgM to IgG) via class switching
24. where the heavy chain constant region is changed by the selection of different heavy chain gene segments.
25. The rest of the heavy chain gene segments which are not selected are permanently deleted.
26. The selected gene segment is joined to the same variable region of heavy chain.

[Total: 25]

L

[Turn over]

- 5 (a) Explain how fluidity of biological membranes can be maintained and the importance of membrane fluidity in biological processes. [10]

[Maintenance of fluidity of biological membranes]

1. Membrane fluidity refers to phospholipids and membrane proteins being always in constant motion / moving laterally in the plane of the membrane.
2. Membrane fluidity can be maintained via amount of unsaturated fatty acids chains / hydrocarbon tails.
3. Greater number of C=C double bonds in the unsaturated fatty acid chains results in more kinks to prevent the close packing of phospholipids, making the membrane more fluid.
4. Membrane fluidity can also maintained by cholesterol.
5. At low temperatures, cholesterol increases membrane fluidity by preventing the close packing of phospholipids.
6. At high temperatures, cholesterol decreases membrane fluidity by interacting with the fatty acid chains of the phospholipid/glycolipid molecules via hydrophobic interactions.

[Importance of membrane fluidity in biological processes]

[Protein secretory pathway]

7. Membrane fluidity is significant to protein transport within the cell / out of the cell / attachment of proteins in the cell surface membrane.
8. **Ref to** membrane-bound vesicles forming and fusing with membrane (of organelle or cell surface membrane).
9. Allows for (transport) vesicles forming / budding from rough endoplasmic reticulum (rER) to Golgi apparatus.
OR
Allows for (transport) vesicles forming / budding from the cis-face to trans-face of Golgi apparatus
OR
Allows for (secretory) vesicles forming / budding from trans-face of Golgi apparatus to cell surface membrane
10. Any named example
[Secretion out of cell] E.g. antibodies by plasma cells, insulin by beta cells via exocytosis
[Attachment of proteins in cell surface membrane] E.g. viral glycoproteins of influenza/HIV virus

[Transport across membrane by bulk transport]

11. Membrane fluidity is significant to endocytosis / phagocytosis / pinocytosis / receptor-mediated endocytosis involving the invagination of cell surface membrane
12. Cell surface membrane to extend outwards / form pseudopodia to surround particles
13. Any named example -
Entry of influenza virus by human respiratory epithelial cells
Entry of pathogens by macrophages / dendritic cells

[Transport across membrane by diffusion or active transport]

14. Membrane fluidity is significant for the formation of transient pores required for simple diffusion of small and non-polar molecules directly across the membrane
15. or to allow for osmosis of water molecules directly across the membrane.

- 16. Allows for transport / channel / carrier proteins to undergo conformational change for facilitated diffusion / active transport of molecules across the membrane.
- 17. Any named example: active transport of H^+ ions by proton pumps / facilitated diffusion of H^+ ions by ATP synthase

[Respiration / photosynthesis]

- 18. Membrane fluidity is significant for oxidative phosphorylation / photophosphorylation to take place.
- 19. **Ref to** conformational change of electron carriers for electrons to be passed down electron carriers of decreasing energy levels

[Cell Signalling]

- 20. Membrane fluidity is significant for cell signalling.
- 21. **Ref to** conformational change of cell surface receptors for activation of receptors / cell signalling pathway.
- 22. Any example: G protein coupled receptor, dimerisation of receptor tyrosine kinase, etc.
- 23. **Ref to** movement of relay proteins along the cell surface membrane for activation of enzymes involved in cell signalling.
- 24. Any example: G protein moves along the cell surface membrane to activate adenylyl cyclase.

[Digestion of substances]

- 25. Membrane fluidity is significant for formation of lysosomes from trans-face of Golgi apparatus.
- 26. Membrane bound lysosome able to fuse with endocytic vesicles / endosomes for autophagy/phagocytosis / digestion / break down of substances.

[Cytokinesis]

- 27. Membrane fluidity is significant for cytokinesis.
- 28. **[Plant cells]** Golgi vesicles to fuse to form the cell plate in plant cells
- 29. **[Animal cells]** Allows for invagination / pinching in of the cell surface membrane in animal cells, forming the cleavage furrow

- (b) Genetic diseases, such as sickle cell anaemia, are caused by gene mutations. Although many of these diseases can be identified through their symptoms, doctors tend to perform further tests in order to accurately diagnose them. These additional tests involve the use of molecular techniques.

Explain how gene mutations can result in sickle cell anaemia, and account for how molecular techniques can be used for the detection of this disease.

[15]

[Gene mutations resulting in sickle cell anaemia]

1. Sickle cell anaemia is an example of a disease caused by a single nucleotide substitution mutation in the gene coding for the β -globin chain in haemoglobin.
2. The thymine nucleotide is substituted / replaced by adenine on the DNA template strand.
3. As a result, the 6th codon in the mRNA is changed from GAG to GUG,
4. and consequently, the 6th amino acid in the polypeptide chain is changed from glutamate / glutamic acid to valine.
5. The R group of glutamate is hydrophilic, whereas that of valine is hydrophobic.
6. This causes the formation of a variant sickle cell haemoglobin (HbS).
7. HbS has an exposed hydrophobic region on the surface of the protein, which is created by the hydrophobic valine residue.
8. As a result, at low oxygen levels, HbS molecules interact with each other via hydrophobic interactions, crystallising into rigid rod-like fibres/structures/shapes.
9. This rod-like fibres of the HbS molecules distort the normal biconcave disc shape of the red blood cell into sickle shape.
10. Blocking blood vessels / sickle red blood cells more fragile.
11. Hence, HbS is less efficient in carrying/transporting oxygen, resulting in sickle cell anaemia.

[Molecular techniques to detect sickle cell anaemia]

12. Sickle cell anaemia is a homozygous recessive disorder.
13. Extract / Obtain DNA samples from the individual in question.
14. Amplify the β -globin gene by polymerase chain reaction (PCR)
15. by designing DNA primers flanking the DNA target sequence.
16. The DNA mixture is heated to about 95°C to denature double-stranded DNA.
17. The DNA mixture is then cooled to about 55°C for DNA primers to anneal to DNA strands.
18. The DNA mixture is then heated to about 72°C for the extension of DNA primers.
19. The normal allele and the mutant allele differs by a single nucleotide.
20. Hence, the mutant allele will have one less restriction site as compared to the normal allele.
21. Digest / Cleave DNA sample with restriction enzyme
22. that cuts at the restriction site which is lost in the mutant allele.

Either

23. Normal allele will be hydrolysed into two restriction fragments with medium and shorter length.
24. Mutant allele will remain as one restriction fragment with longer length.

OR

23. Formation of different number of restriction fragments and
24. different lengths of restriction fragments in the two different alleles.

25. Run gel electrophoresis to separate the DNA fragments by size.
26. The negatively-charged DNA molecules will move towards the positive end of the gel / anode.
27. Different DNA banding patterns will be produced by the two different alleles,
28. as the smaller DNA molecules will move further away from the wells / as the larger DNA molecules will be found nearer to the wells.
29. Perform Southern blotting by designing a radioactively labelled probe
30. with nucleotide sequence complementary to sequence of all three restriction fragments.
31. Homozygous dominant, heterozygote/carrier and homozygous recessive/affected individuals will have different DNA banding patterns.
32. An individual with sickle cell anaemia will have one DNA band found nearer to the well.
33. Thicker bands as individual has two copies of the same / mutant allele.
34. Hence, each DNA band will contain twice the amount of DNA as compared to the other bands,
35. and each DNA band will hybridise to twice the amount of radioactive probe.
36. A carrier/heterozygote will have three DNA bands
37. as individual has one copy of the normal allele, and one copy of the mutant allele.
38. A homozygous dominant individual will have two DNA bands found further away from the well.
39. Thicker bands as individual has two copies of the same/normal allele.

[Total: 25]